

My choice, my actions: self-determination, not instrumental value of outcomes enhances outcome monitoring during learning

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Freedom of choice enhances our sense of agency. During goal-directed behavior, the freedom to choose between different response options increases the neural processing of positive and negative feedback, indicating enhanced outcome monitoring under conditions of high agency experience. However, it is unclear whether this enhancement is predominantly driven by an increased salience of self- compared to externally determined action outcomes or whether differences in the perceived instrumental value of outcomes contribute to outcome monitoring in goal-directed tasks. To test this, we recorded electroencephalography while participants performed a reinforcement learning task involving free choices, action-relevant forced choices, and action-irrelevant forced choices. We observed larger midfrontal theta power and N100 amplitudes for feedback following free choices compared with action-relevant and action-irrelevant forced choices. In addition, a Reward Positivity was only present for free but not forced choice outcomes. Crucially, our results indicate that enhanced outcome processing is not driven by the relevance of outcomes for future actions but rather stems from the association of outcomes with recent self-determined choice. Our findings highlight the pivotal role of self-determination in tracking the consequences of our actions and contribute to an understanding of the cognitive processes underlying the choice-induced facilitation in outcome monitoring.

Key words: electroencephalography; feedback processing; freedom of choice; outcome relevance; sense of agency.

Introduction

When interacting with the environment, we sometimes feel more or less in control over our actions and their effects. This experience of control reflects our sense of agency and serves as a cue to distinguish self- from externally determined actions (Haggard 2017). In conjunction with supporting a self–other differentiation, sense of agency facilitates an adaptive interaction with the environment (Ly et al. 2019; Studer et al. 2020). We can instrumentalize the knowledge that we are the agent of our actions to adopt behavioral strategies that promote goal attainment (Luo et al. 2022). Consequently, sense of agency is crucial for a coherent sense of self- and goal-directed behavior.

Agency experience is influenced by the integration of sensory, cognitive, and affective cues that operate prior (e.g. causal beliefs), during (e.g. sensorimotor signals), and after an action (e.g. contingency of action effects; Moore et al. 2009; Synofzik et al. 2013; Sidarus et al. 2017). A large body of work has focused on studying how the sense of agency affects the processing of action outcomes using electroencephalography (EEG; see Gentsch and Schütz-Bosbach 2015; Moore 2016; Giersiepen et al. 2023). In these studies, participants' sense of control over auditory or visual action effects is typically manipulated by varying their motor involvement or their degree of choice over response options and differences in outcome processing are examined. Results indicate a neural differentiation of self- and externally induced action effects already during early sensory stages of feedback processing,

as reflected in the N100 component, a negativity in the event-related potential (ERP) waveform peaking around 100 ms after feedback onset (e.g. Martikainen et al. 2005; Gentsch and Schütz-Bosbach 2011; Weiss et al. 2011; Kaiser and Schütz-Bosbach 2018). Most of these studies report a reduction in N100 amplitude for action outcomes produced under conditions of high compared to low sense of agency, a phenomenon referred to as sensory attenuation (see Gentsch and Schütz-Bosbach 2015). Crucially, in the reported studies, action outcomes like visual stimuli or sounds typically serve as sensory confirmation of a completed action, without any instrumental value for future actions. Hence, they are not associated with instrumental positive or negative affective value indicative of participants' success or failure in attaining an action goal (see Haggard and Eitam 2015).

Interestingly, previous work indicates that the attenuation effect is reversed if action outcomes under conditions of high and low sense of agency are equally predictable (Kaiser and Schütz-Bosbach 2018). Furthermore, affective processing has been proposed to be intricately linked to the sense of agency (Synofzik et al. 2013; Gentsch and Synofzik 2014; Kaiser et al. 2021b; Majchrowicz and Wierchoń 2021). Thus, early processing of action outcomes, as reflected in the visual N100, may be sensitive to both the affective context and the predictability of task-relevant action effects.

During goal-directed behavior, affective outcomes are informative cues that reinforce or inhibit previously performed actions.

Thus, affective feedback carries practical significance, reflecting current task performance and serving as an instructive cue to optimize behavior. Despite the central role of the sense of agency in successful goal-directed behavior and the apparent importance of affective feedback for an adaptive interaction with the environment (Synofzik et al. 2013; Gentsch et al. 2015; Kaiser et al. 2021b), relatively little work has focused on studying how the experience of agency, particularly through self-determined choice, influences outcome processing in goal-directed and affective contexts. Using random reward schedules, involving equal occurrences of gains and losses, controls feedback processing differences that might result from frequency effects, such as those related to prediction errors (e.g. Holroyd et al. 2003). However, random reward schedules also preclude the study of how agency experience influences the processing of instrumental affective feedback during contexts, where participants can learn to optimize their behavior according to nonrandom reward contingencies.

Some studies have shown that affective context, in the sense of whether action outcomes carry positive or negative instrumental value, impacts how the sense of agency influences the processing of these outcomes (Gentsch et al. 2015; Christensen et al. 2016; Majchrowicz and Wierzbicki 2021). The results indicate a differentiation of positive and negative action outcomes as reflected in self-reports and implicit measures, such as the N100 component. Specifically, individuals are more inclined to experience agency over positive rather than negative action outcomes, which has been interpreted as a self-serving attribution bias (e.g. Gentsch et al. 2015; Chambon et al. 2020).

In addition to sensory processing, as reflected in the N100 component, affective contexts involve the evaluation of outcome value (Kaiser et al. 2021b). Negative compared to positive feedback is typically associated with greater midfrontal theta (MF0; 4 to 7 Hz) power (Cavanagh et al. 2010; Van de Vijver et al. 2011; Kaiser et al. 2021a; Giersiepen et al. 2023). MF0 oscillations are assumed to reflect cognitive control and cognitive conflict during value-based decision-making (Cavanagh and Frank 2014; Weismüller et al. 2019; Kaiser et al. 2022). Conversely, positive feedback has been shown to elicit larger voltage deflections compared to negative feedback over similar midfrontal scalp sites, resulting in a Reward Positivity (RewP) that peaks 250 to 350 ms after feedback onset (Holroyd et al. 2008; Proudfit 2015; Zheng et al. 2020). Source analyses indicate that both feedback-induced MF0 oscillations and RewP originate from the anterior cingulate cortex (ACC; see Cavanagh and Frank 2014). The ACC is highly interconnected with other cortical and subcortical areas and is considered functionally relevant for behavioral monitoring and adaptation (Cavanagh and Frank 2014; Monosov et al. 2020; Kaiser et al. 2023; Qu et al. 2023). Importantly, valence-specific modulations of RewP and MF0 power seem to serve a functional role during feedback learning (Cohen and Donner 2013; Williams et al. 2018).

Studies investigating how action choice impacts affective outcome processing usually find an increased RewP amplitude and MF0 power for free choices (associated with high sense of agency) compared to forced choices (associated with low sense of agency; Hassall et al. 2019; Zheng et al. 2020; Giersiepen et al. 2023). Likewise, several studies indicate heightened outcome monitoring during active compared to observational learning (Bellebaum et al. 2010; Weismüller et al. 2019). Thus, self-determined actions may enhance the neurophysiological processes associated with outcome evaluation during goal-directed behavior. Importantly, freedom of choice seems to increase the monitoring of positive and negative action outcomes to a similar extent (Weismüller et al. 2019; Giersiepen et al. 2023), indicating the absence of a

valence-specific (self-serving) bias during later stages of feedback processing (but see Zheng et al. 2020).

Crucially, the results from these studies do not unequivocally isolate the cognitive processes involved in the neural enhancement of outcomes during self-determined choice. On the one hand, high compared to low sense of agency during free compared to externally determined choices may be associated with a greater perceived *action-relevance* of affective feedback, meaning that participants monitor outcomes resulting from free compared to forced choices more strongly as they carry information with practical relevance for subsequent action selection decisions. For example, negative feedback during free-choice trials might inform participants not to repeat the same choice in subsequent trials. In this case, the *instrumental value*, i.e. the perceived capacity to use performance feedback to adjust behavior, would be the main factor leading to enhanced outcome processing. On the other hand, freedom of choice may induce a general increase in task engagement, independent of whether outcomes are *action-relevant* or *action-irrelevant* for future behavior, potentially reflecting a bias toward weighting self-induced action outcomes as more salient, even when outcomes of both free- and forced-choice conditions are equally informative of participants' goal achievement (Deci and Ryan 2012; Gentsch et al. 2015; Mühlberger et al. 2017; Chambon et al. 2020). For example, during reinforcement learning, humans may value negative feedback during free choices more than negative feedback during forced choices because they directly relate this value to a recent self-determined task choice.

With the goal of isolating the cognitive processes underlying the choice-induced enhancement in affective outcome monitoring observed in previous studies, the current study differentiates the neurocognitive effects of self-determined choice on the one hand and outcome relevance on the other hand in a motivationally salient, goal-directed task. To this end, we recorded EEG and manipulated participants' agency experience by varying their freedom of choice in a reinforcement learning task employing probabilistic item–outcome associations and monetary gains and losses as affective action effects. By monitoring action outcomes, participants could learn which of the two items constituted the high-value item (i.e. item choice likely results in a monetary gain) and which constituted the low-value item (i.e. item choice likely results in a monetary loss).

To examine how the sense of agency, operationalized as freedom of choice, influences different stages of outcome evaluation, we assessed both early and (N100) and later markers (RewP, MF0) of feedback processing. We expected a significant increase in outcome monitoring for free choices compared with forced choices, reflected in larger ERP deflections and time–frequency power during feedback presentation. Furthermore, we predicted an RewP in response to positive feedback and a larger increase in MF0 power in response to negative feedback.

Crucially, unlike previous studies (e.g. Weismüller et al. 2019; Chang et al. 2020; Zheng et al. 2020; Giersiepen et al. 2023), we separated the impact of outcome relevance and choice autonomy by establishing three choice scenarios: (1) *free-choice* blocks, where participants could choose any of two items in each trial, (2) *action-irrelevant forced-choice* blocks, where participants confirmed a pre-set choice without using feedback for decision-making, and (3) *action-relevant forced-choice* blocks with forced choices initially, followed by free choices, allowing the initial feedback to inform later decisions. This approach tests whether the choice-induced increase in feedback processing is due to its perceived importance for future actions or a general bias beyond the immediate relevance of feedback for future action decisions.

We anticipated that these effects would be paralleled by higher self-reported agency experience when participants were given the freedom to choose between response options, compared to being confined to selecting a predetermined option.

Materials and methods

Participants

Inclusion criteria for this study were an age between 18 and 45 years, normal or corrected-to-normal visual acuity without color vision deficiencies, and the absence of an acute neurological or psychiatric disorder. We collected data from 40 participants. During data analysis, three participants were excluded (see [Data Preprocessing](#)). The final test set thus comprised 37 participants ($M_{\text{age}} = 25.5$ years, $SD = 4.9$, 20 female, 16 male, 1 gender not indicated, 33 right-handed). Power analysis and comparisons with previous EEG studies examining choice-induced within-subject differences in outcome processing (e.g. [Mühlberger et al. 2017](#); [Giersiepen et al. 2023](#)) were conducted to ensure adequacy in detecting the effects of choice and outcome valence on feedback processing. Based on these insights, we concluded that our sample size is likely suitable to detect such effects. Further details are provided in the [Supplementary Materials](#). Participants received a financial reimbursement of 9 Euro per hour or participation credit. Independent of compensation choice, participants received a performance-dependent monetary bonus at the end of the experiment ($M = 9.64$ Euro, $SD = 1.03$, range: 6.90 to 10.86; see [Materials and Procedure](#)). Study approval was obtained from the ethical board of the Department of Psychology at the Ludwig-Maximilians-University Munich. Participants voluntarily participated in the study, and written consent was obtained at the beginning of the experimental session.

Apparatus and measurement setup

We employed 64 active electrodes embedded in an elastic cap for EEG signal acquisition (actiCAP snap, Brain Products, Munich, Germany). The electrodes were placed in line with the extended version of the international 10-20 system. Electrode FCz served as an online reference, and a ground electrode was placed at location FPz. We employed a BrainAmp DC amplifier (BrainVision) for signal amplification, and the continuous signal was digitized at a sampling rate of 1000 Hz. The data were filtered online using a bandpass filter with half-amplitude cutoffs of 0.016 and 250 Hz. Electrode-to-skin impedances were kept below 20 k Ω .

Materials and procedure

Participants were seated approximately 70 cm from a 24-inch screen on which a reinforcement learning task was presented. MATLAB R2020B and Psychophysics Toolbox extensions (Psychtoolbox Version 3; [Brainard and Vision 1997](#)) were used to implement the task.

In the task, participants made binary item selections on each trial. Choice options comprised two differently colored squares presented horizontally aligned adjacent to the screen center. Unbeknown to participants, one of the two options was associated with a 75% probability of gaining 4 cents (high-value item), while the other option was associated with a 75% probability of losing 2 cents (low-value item; [Tversky and Kahneman 1992](#)). Whether an item choice resulted in a gain or loss of money was presented as affective action effect after item selection. The task was performed in separate blocks, each of which used two novel, differently colored squares as targets. Thus, each block necessitated the learning of novel stimulus-reward contingencies,

and participants could learn from feedback after every trial which of the two items was more likely to lead to financial gain or loss.

To manipulate agency experience, choice context varied between three different types of blocks: free-choice blocks, action-irrelevant forced-choice blocks, and action-relevant forced-choice blocks. During free-choice blocks, participants were free to choose either of the two stimuli presented on the screen by pressing the left or right arrow keys on the keyboard. During action-irrelevant forced-choice blocks, item selections were determined by the computer throughout the entire block, thereby withdrawing participants' control over item choices and their ensuing gains and losses. In action-relevant forced-choice blocks, participants were instructed to first complete 54 (75%) forced-choice trials, followed by 18 free choices during the final quarter of trials. Thus, during action-relevant forced-choice blocks, but not during action-irrelevant forced-choice blocks, feedback during forced-choice trials was informative for participants' future behavior. Prior to each block, participants were informed if during the upcoming block they would always (free-choice blocks), never (action-irrelevant forced-choice blocks), or only in the latter part of the block (action-relevant forced-choice blocks) be free to choose which item to select. To match motor involvement during free- and forced-choice trials, participants confirmed computer choices with button presses during forced choices (see also [Chambon et al. 2020](#)).

Trials of all conditions started with a fixation cross presented in the screen center (jittered for 1.5 to 1.8 s). Next, participants performed item selections (1 s), followed by a choice confirmation period (1 s). Finally, "+4" or "-2" were presented as affective action effect during the feedback period (1 s; [Fig. 1](#)). All stimuli were presented on a gray background with a visual angle of 1.2°. Participants were instructed to use their right index finger for item selection. In cases where participants did not respond during the choice period, the screen did not display gain or loss feedback, but instead a message prompting them to respond faster appeared. If they tried to select the item not chosen by the computer, participants were informed that they had pressed the wrong key. Trial durations ranged from 3.5 to 4.8 s.

The task comprised four blocks per choice condition (12 blocks in total), each consisting of 72 trials (864 trials total; 288 trials per choice condition). The 24 targets (two target colors per block) were determined to be maximally perceptually distinct from each other and the background according to the CIELAB color space ([Holy 2022](#); [Giersiepen et al. 2023](#)). Which of the two items corresponded to the high- and low-value item was determined randomly at the beginning of each block. Stimuli appeared equally often and in random order on the left and right side of the screen center. Item selections during forced choices from action-irrelevant and action-relevant forced-choice blocks were yoked to participants' responses during previous free-choice blocks. Consequently, all participants started in the free-choice condition, followed by the action-irrelevant and the action-relevant forced-choice condition. This was necessary to allow for an equivalent sequence and frequency of high- and low-value item choices in all three choice conditions. Participants were informed about their block-wise monetary gain at the end of each block. Importantly, as participants were informed prior to the experiment, they received the money gained during their two best blocks of each condition after the experiment, making choice-related gains and losses relevant to participants in all choice conditions.

To measure the effect of our choice manipulation on agency experience, we obtained participants' agency ratings at the end of each block. In line with previous work ([Giersiepen et al. 2023](#)),

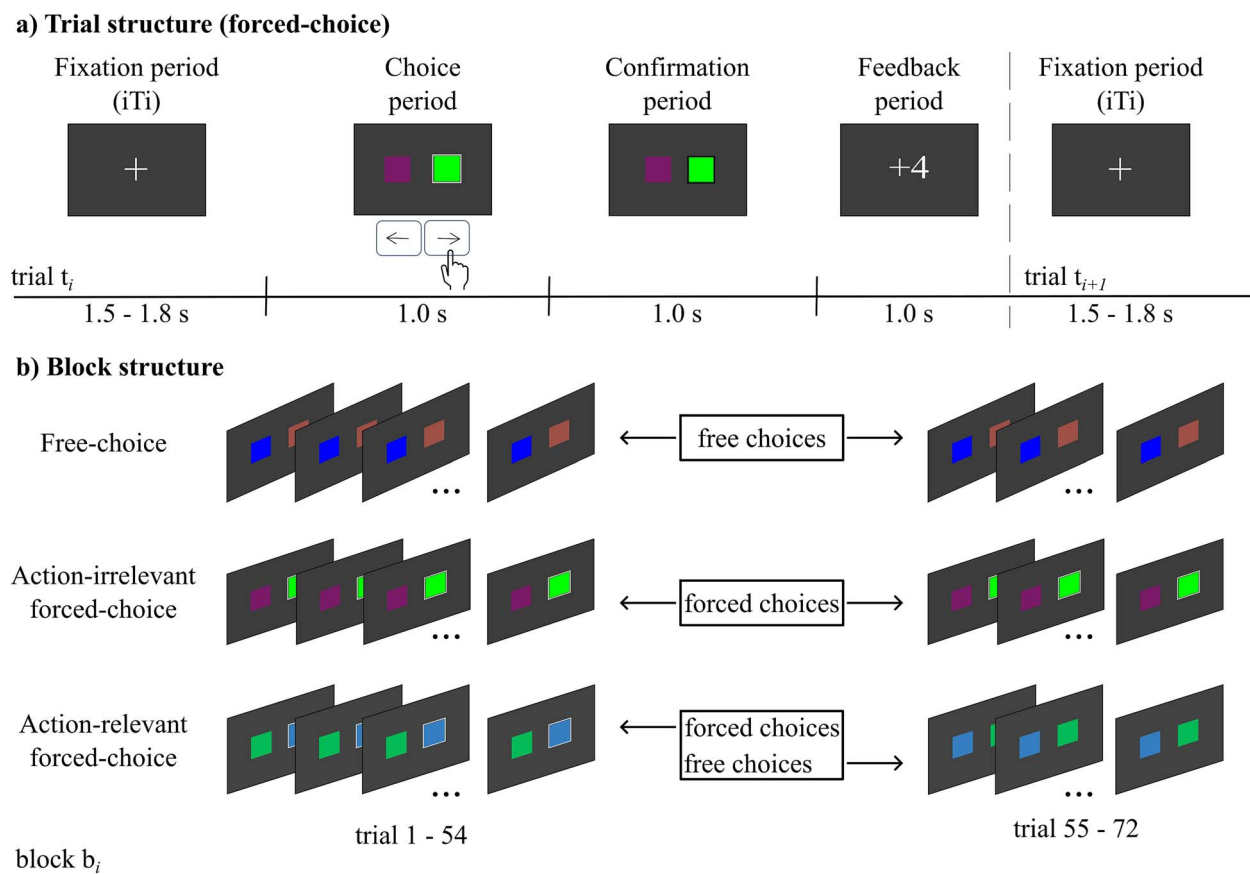


Fig. 1. Schematic representation of the experimental task. a) Trial structure exemplified by a forced-choice trial. Each trial comprised an item choice followed by a black rectangle around the selected item to confirm participants' key presses. Probabilistic affective feedback was presented as action effect. Trials were separated by a white fixation cross presented at the screen center. b) Block structure depicting examples of free-choice (upper row), action-irrelevant forced-choice (middle row), and action-relevant forced-choice (bottom row) blocks. During free-choice blocks, participants freely selected one of the two items. During action-irrelevant forced-choice blocks, the computer selected one item in all trials, signaled by a white frame around the selected stimulus. Item choices had to be confirmed by participants. During action-relevant forced-choice blocks, participants performed forced-choices during the first 75% of trials (trial 1 to 54) and could freely choose between items during the final 25% of trials (trial 55 to 72). Each block employed a novel pair of colored squares as targets.

participants were asked to indicate their agreement with two statements (item 1: "I was free to choose any item I wanted in this block."; item 2: "I was in control over the amount of money I gained in this block.") on a bipolar quasi-continuous scale, ranging from *Strongly Disagree* to *Strongly Agree*. As highlighted in theoretical work (Pacherie 2008) and empirical evidence from previous studies (Sidarus et al. 2017; Giersiepen et al. 2023), the sense of agency is a multidimensional construct influenced by different aspects related to an action. Consequently, we separately evaluated individuals' perceived control over their item choices (item 1: choice agency) and the outcomes of these choices (item 2: outcome agency; Kaiser et al. 2021b). After agency ratings, participants could advance to the next block at their own pace.

Participants completed a practice session of blocks of 10 free-choice, action-irrelevant forced-choice, and action-relevant forced-choice trials to familiarize themselves with the task. The entire experimental session took approximately 2.5 hours per participant.

Data analysis

Behavioral analysis

MATLAB R2020B and RStudio (Version 2023 June 1) were used for behavioral data analyses. To evaluate participants' learning performance, we tested whether the proportion of high-value item

choices was significantly higher than chance level during free choices performed during free-choice and action-relevant forced-choice blocks using a one-sample t-test (two-tailed). Performance was examined by plotting participants' cumulative proportion of high-value item choices over bins of four trials (19 bins) during free-choice blocks, action-relevant forced-choice blocks, and action-irrelevant forced-choice blocks.

We also tested whether participants exhibited different reaction times for high-value item choices in free-choice blocks, forced-choice trials from action-relevant forced-choice blocks, and forced-choice trials from action-irrelevant forced-choice blocks using paired samples t-tests (two-sided).

Repeated-measures ANOVAs with the factors CHOICE (free-choice/action-relevant forced-choice/action-irrelevant forced-choice) and RATING (choice agency/outcome agency) were conducted to assess whether our freedom of choice manipulation influenced participants' postblock agency ratings. To examine the interaction of CHOICE and RATING, follow-up contrasts (t-test, two-sided) were performed, testing whether the effect of choice was larger for choice agency or outcome agency ratings. We quantified participants' judgments as percentage agreement with statements 1 and 2 (*Strongly Disagree*=0%, *Strongly Agree*=100%; Giersiepen et al. 2023). One participant did not provide agency ratings for the last action-relevant

forced-choice block. Here, agency experience was quantified using the mean values of the remaining three action-relevant forced-choice blocks. Agency judgments revealed eight extreme outliers (i.e. values $< [1\text{st quartile} - 3 * \text{interquartile range}]$ or values $> [3\text{rd quartile} + 3 * \text{interquartile range}]$). Because the outliers did not change the direction or significance of the results, the reported statistics are based on the dataset including extreme values. Follow-up test results were corrected for multiple comparisons using Holm's (1979) method. In cases where Mauchly's test indicated violation of sphericity, we performed Greenhouse–Geisser corrections (uncorrected df s are reported for all tests).

Effect sizes for condition-wise differences are reported as Cohen's d (small effect: $d = 0.20$; medium effect: $d = 0.50$; large effect: $d = 0.80$), partial eta-squared (small effect: $\eta_p^2 = 0.01$; medium effect: $\eta_p^2 = 0.06$; large effect: $\eta_p^2 = 0.14$; Cohen 1988). Bayes factors (BF) are presented to quantify the evidence for the null (H_0), as well as the alternative hypothesis (H_1 ; weak evidence for H_1 : $BF_{10} = 1$ to 3; positive evidence for H_1 : $BF_{10} = 3$ to 20; strong evidence for H_1 : $BF_{10} = 20$ to 150; very strong evidence for H_1 : $BF_{10} > 150$; Raftery 1995).

Data preprocessing

Offline analyses were performed in MATLAB R2020B using Field-Trip software toolbox (Oostenveld et al. 2011). First, a Butterworth Infinite Impulse response two-pass filter (roll-off: 6 db/octave), using 1.0 and 40 Hz as low and high cut-offs, was applied to the raw data. Next, the average of all active electrodes was defined as offline reference (Kaiser et al. 2021a; Giersiepen et al. 2023). Epochs were created from 3,500 ms prior to and 2,500 ms after feedback onset. Next, the data were downsampled to 250 Hz and artifacts resulting from eyeblinks and saccadic eye movements were identified using independent component analysis (logistic infomax ICA algorithm *runica* as implemented in FieldTrip). An average of 2.9 components ($SD = 0.9$, range: 1 to 4) were removed, and the remaining components were projected back to the data. Activation patterns were baseline corrected by subtracting the grand average activity of 200 ms before feedback onset for every trial.

Trials with substantial noise influence (i.e. voltage deflections greater than $\pm 100 \mu V$) were subsequently removed from each dataset, leading to the exclusion of 6.0% trials ($SD = 5.1$, range: 0 to 17.4). We further removed the data of one to three noisy electrodes for 21 participants ($M = 1.1$ electrodes, $SD = 0.5$). Activity at the channel level was reconstructed using spherical spline interpolation (see also Perrin et al. 1989). For condition-wise statistical analyses, only valid trials, defined as trials in which one of the two items was selected during the choice period of free-choice trials or trials in which participants confirmed computer choices during forced-choice trials, were retained, leading to a mean removal of 18.6 invalid trials ($SD = 22.3$, range: 3 to 139) per participant. Note that action-relevant forced-choice blocks ended with a small number of free-choice trials. These were only included to allow us to investigate whether the anticipation of free choices would modulate the processing of feedback presented in response to preceding forced-choice trials. Accordingly, only forced-choice trials of action-relevant forced-choice blocks were included in statistical analyses. Finally, because reinforcement learning is associated with a strong decrease in low-value item choices over time (cf. Fig. 2), we focused our analysis on high-value item choices. Specifically, unlike designs with random reward schedules, learnable tasks with relatively stable reward contingencies, as in the current study, contain very few low-value item

Table 1. Mean number of trials (and standard deviations) for all choice-outcome conditions relevant to electrophysiological statistical analyses.

Trial type	Feedback valence	
	Negative	Positive
Free-choice	58.6 trials (8.3)	172.1 trials (23.1)
Action-irrelevant forced-choice	59.6 trials (6.2)	172.8 trials (20.1)
Action-relevant forced-choice	42.3 trials (6.9)	125.9 (17.8)

choices, especially those with positive feedback, making the analysis of these trials potentially unreliable and highly noisy.

Owing to noncompliance with the task (i.e. selecting the item not chosen by the computer > 2.5 SD more often than average during forced choices), absence of learning (i.e. selecting the low-value item > 2.5 SD more often than average during free choices), or fewer than 25 trials in at least one of the experimental conditions after preprocessing, we excluded three participants from the dataset, resulting in a final sample of 37 participants. In the final dataset, no significant differences in the number of errors could be detected between the free-choice condition ($M = 5.80$, $SD = 3.67$), the action-relevant forced-choice condition ($M = 3.84$, $SD = 4.25$), and the action-irrelevant forced-choice condition ($M = 5.22$, $SD = 4.25$, $F(2, 72) = 2.82$, $P = 0.080$, $\eta_p^2 = 0.07$, $BF_{10} = 0.82$). Participants completed an average of 631.35 trials ($SD = 73.2$, range: 480 to 750; Table 1).

Time-frequency analysis

We used a Surface Laplacian filter (polynomial degree: 10; Perrin et al. 1989; Cohen 2014) to increase the topographical specificity of our time-frequency results. To identify the frequency power over time, we subsequently performed Morlet wavelet convolution. During convolution, the number of cycles linearly increased from three to eight in 19 steps, encompassing frequencies from 2 to 20 Hz (Cohen 2014; Giersiepen et al. 2023). Epochs for statistical analyses were cut from $-1,500$ to $1,000$ ms around feedback onset. The time window from -300 to -100 ms was used for time-frequency-specific baseline normalization. To this end, we converted the data to decibel scale ($dB = 10 * \log_{10}[\text{power}/\text{baseline}]$) and corrected all time-frequency points by the average oscillatory power over all trials.

Event-related potentials

We employed a time window ranging from -200 to $1,000$ ms relative to feedback onset to visualize ERP waveforms during feedback presentation.

Statistical analyses

In line with previous studies, the effect of CHOICE (free-choice/action-relevant forced-choice/action-irrelevant forced-choice) and feedback VALENCE (negative feedback/positive feedback) on MF0 power was expected to be most pronounced from 200 to 600 ms after feedback onset (e.g. Zheng et al. 2020; Kaiser et al. 2022; Giersiepen et al. 2023). To account for temporal smoothing inherent to time-frequency data (Cohen 2014), significant condition differences in midfrontal oscillatory activity were examined from 0 to $1,000$ ms relative to feedback onset. We expected the N100 and RewP components to peak around 100 and 300 ms postfeedback onset, respectively (Weiss

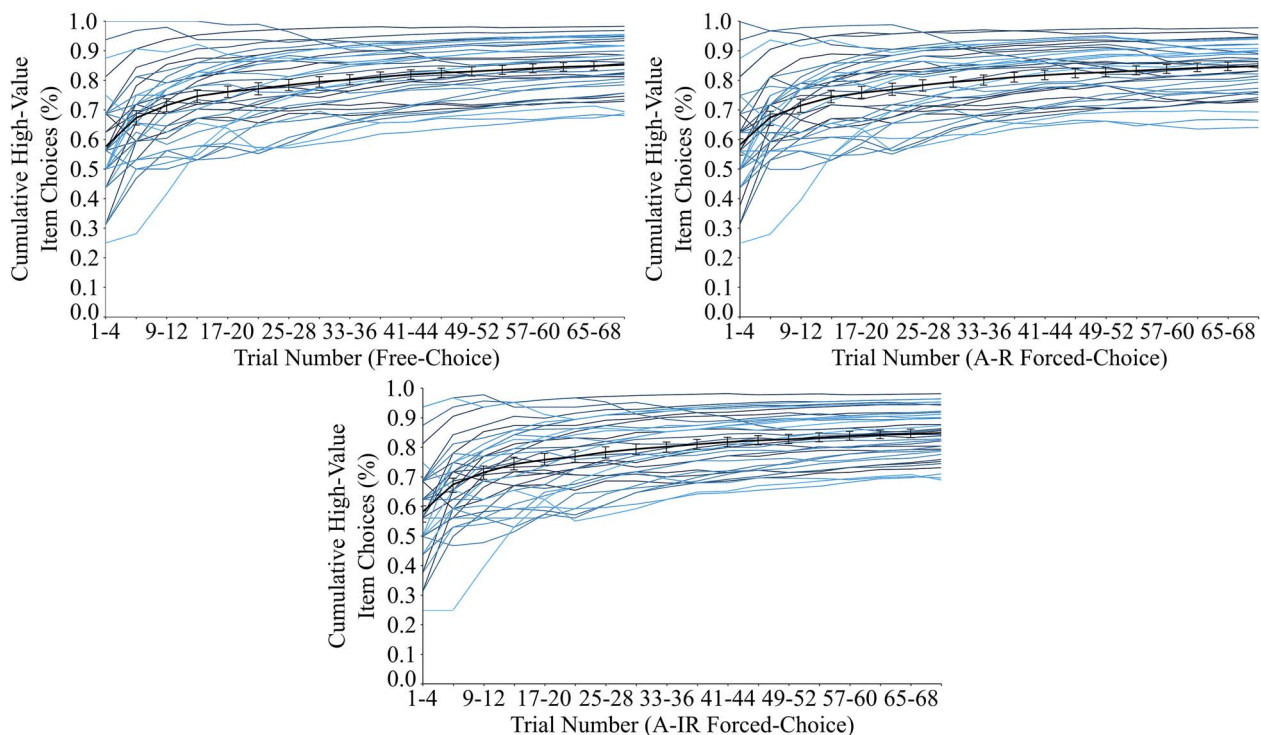


Fig. 2. Behavioral results depicting participant performance as the cumulative proportion of high-value item choices for free-choice blocks (upper left), action-relevant (A-R) forced-choice blocks (upper right), and action-irrelevant (A-IR) forced-choice blocks (lower middle). Black lines show participants' grand average performance, including standard errors. Note that forced-choice item selections were manipulated to mirror participants' free choices.

et al. 2011; Proudfit 2015; Kaiser and Schütz-Bosbach 2018; Zheng et al. 2020). We thus examined differences associated with CHOICE and VALENCE during the first 500 ms of feedback presentation, encompassing both ERP components of interest.

Both the theta oscillations and the RewP were previously shown to be most pronounced over midfrontal scalp sites. In addition, agency experience has consistently been found to modulate the N100 component across frontal, central, and parietal scalp sites (e.g. Gentsch and Schütz-Bosbach 2011; Hughes and Waszak 2011; Gentsch et al. 2015; Weismüller et al. 2019). To allow for direct comparison, we therefore employed the same midfrontal electrodes for time-frequency and ERP data and averaged activity for statistical analyses over the midfrontal electrodes Fz, FCz, FC1, and FC2.

We tested the effect of the factors CHOICE and VALENCE on midfrontal oscillatory activity and feedback-locked ERPs using cluster-based permutation *F*-tests and *t*-tests (two-sided, $\alpha = 0.05$) as implemented in FieldTrip (Oostenveld et al. 2011). This approach circumvents the multiple comparison problem inherent to the analysis of multidimensional EEG data (Maris and Oostenveld 2007). During permutation analyses, significant differences of individual data points are evaluated against a predefined alpha level and data points adjacent in time and frequency are summarized into clusters. Cluster size, defined as the sum of test statistics (i.e. *F*- or *t*-values) of all significantly different adjacent data points, is used to determine whether significant differences between conditions exist. Random permutations of the original data are drawn to evaluate whether the probability of finding a cluster of the observed size is greater than would be expected by chance.

We applied the Monte Carlo Method to the largest of the clusters and performed 1,000 random permutations to evaluate

whether the data of our experimental conditions stem from the same probability distributions (Maris and Oostenveld 2007). During each permutation, the condition labels, indicating to which condition each data point belongs, were shuffled randomly, while keeping the data points themselves fixed. This process created a null distribution of the test statistic by breaking any systematic relationship between the conditions and the observed data, allowing for the computation of a *P*-value that reflects the likelihood of observing the test statistic if no significant condition differences exist.

Crucially, while the resulting clusters can indicate statistically significant differences between conditions, they do not specify the exact time windows, frequencies, and electrodes where these differences occur (Sassenhagen and Draschkow 2019). Therefore, we confined our analyses to the prespecified time windows and regions of interest (Maris and Oostenveld 2007) and report statistical inferences from cluster-based permutation results along with descriptive information about the cluster's features.

To test whether freedom of choice differentially affects the processing of negative and positive feedback (CHOICE $\times$ VALENCE interaction), we subtracted the mean power values during positive feedback presentation from the mean power values of negative feedback for MF0 analyses, as well as mean voltage deflections during negative feedback from the activity observed during the presentation of positive feedback for the N100 and RewP. The resulting condition-wise differences were entered in a repeated-measures ANOVA. To extract the RewP from other ERP components, the resulting difference waveform (positive – negative feedback) was plotted (Proudfit 2015).

Cohen's *d* is reported as a measure of effect size. Effect sizes for significant differences between conditions were determined based on the time (and frequency) characteristics of the clusters detected in the data.

Results

Behavioral results

Participant performance depicting the cumulative proportion of high-value item choices throughout free-choice blocks, action-relevant forced-choice blocks, and action-irrelevant forced-choice blocks is displayed in Fig. 2.

The high-value item was selected significantly more often than chance level during free-choice blocks ($M = 86.9\%$, $SD = 8.5$, range: 69.9 to 99.0), $t(36) = 24.74$, $P < 0.001$, CI 95% [237.2, 253.8], Cohen's $d = 9.83$, $BF_{10} > 10^6$. Participants also selected the high-value item significantly more often than chance level ($M = 86.9\%$, $SD = 10.6$, range: 59.7 to 1.0) in free-choice trials of action-relevant forced-choice blocks, $t(36) = 18.98$, $P < 0.001$, CI 95% [58.3, 63.7], Cohen's $d = 7.62$, $BF_{10} > 10^6$. Thus, participants successfully learned item-reward contingencies in free-choice, and action-relevant forced-choice blocks.

Reaction times of trials during free-choice blocks were significantly higher ($M = 519.67$ ms, $SD = 62.54$, range: 409.63 to 681.26) than reaction times of trials during action-irrelevant forced-choice blocks ($M = 508.22$ ms, $SD = 68.10$, range: 392.84 to 690.22), $t(36) = 2.84$, $P = 0.007$, CI 95% [3.27, 19.64], Cohen's $d = 0.17$, $BF_{10} = 5.41$. Also, reaction times during free-choice blocks were significantly higher than reaction times in action-relevant forced-choice blocks ($M = 508.85$ ms, $SD = 62.92$, range: 420.70 to 691.37), $t(36) = 2.31$, $P = 0.027$, CI 95% [1.32, 20.32], Cohen's $d = 0.17$, $BF_{10} = 1.83$. No significant differences in reaction times were detected for action-relevant forced-choice blocks and action-irrelevant forced-choice blocks, $t(36) = 0.20$, $P = 0.841$, CI 95% [-5.74, 7.02], Cohen's $d = 0.01$, $BF_{10} = 0.18$. Thus, item selections during free choices involved longer response times than those requiring the execution of predetermined responses. This might be because, during forced-choice trials, participants could immediately confirm the preselected item choices without the need to consider the optimal choice. Supporting this interpretation, we also found that forced choices compared to free choices, were associated with higher motor readiness potentials (see [Supplementary Materials](#)).

Regarding participants' self-reported agency experience, a two-way ANOVA with the factors CHOICE (free-choice/action-relevant forced-choice/action-irrelevant forced-choice) and RATING (choice/outcome) showed a main effect of CHOICE, with highest judgments for free-choice blocks ($M = 71.5\%$, $SD = 19.3$), followed by action-relevant forced-choice blocks ($M = 41.4\%$, $SD = 8.7$) and action-irrelevant forced-choice blocks, $M = 12.3\%$, $SD = 12.3$, $F(2, 72) = 191.22$, $P < 0.001$, $\eta_p^2 = 0.84$, $BF_{10} > 10^6$ (Fig. 3). In addition, a main effect of RATING was observed, with higher scores for the statement pertaining to freedom of choice ("I was free to choose any item I wanted in this block."; $M = 46.4\%$, $SD = 12.4$) than outcome controllability ("I was in control over the amount of money I gained in this block."), $M = 37.1\%$, $SD = 13.3$, $F(1, 36) = 17.25$, $P < 0.001$, $\eta_p^2 = 0.32$, $BF_{10} = 1.52$. A significant CHOICE $\times$ RATING interaction indicated that the effect of choice depended on rating item, meaning that choice condition differentially affected participants' ratings pertaining to choice agency and outcome agency, $F(2, 72) = 29.11$, $P < 0.001$, $\eta_p^2 = 0.45$, $BF_{10} > 10^6$. Post-hoc contrasts revealed that free-choice blocks led to significantly higher choice agency ratings and outcome agency ratings than action-relevant forced-choice blocks (choice agency: $\Delta M = 42.8\%$, $t(36) = 13.58$, $P < 0.001$, CI 95% [36.73, 49.15], Cohen's $d = 2.56$, $BF_{10} > 10^6$; outcome agency: $\Delta M = 17.4\%$, $t(36) = 4.41$, $P < 0.001$, CI 95% [9.39, 25.34], Cohen's $d = 0.74$, $BF_{10} = 276.66$), and action-irrelevant forced-choice blocks

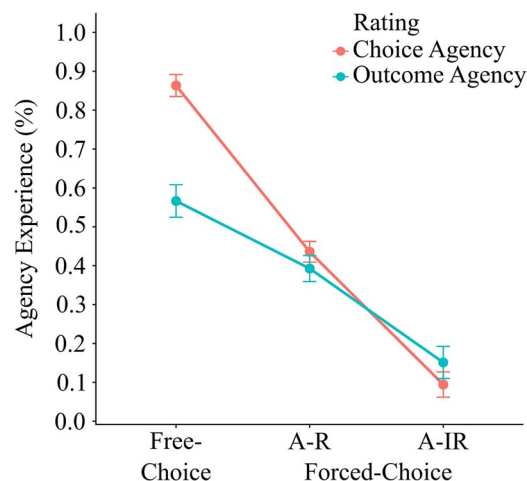


Fig. 3. Mean agency experience (including standard errors) in the three choice conditions (free-choice/action-relevant [A-R] forced-choice/action-irrelevant [A-IR] forced-choice), averaged over participants and blocks. Mean values and lines separately depict participants' agreement with statements relating to freedom of choice (choice agency: "I was free to choose any item I wanted in this block.") and outcome controllability (outcome agency: "I was in control over the amount of money I gained in this block.").

(choice agency: $\Delta M = 76.9\%$, $t(36) = 18.31$, $P < 0.001$, CI 95% [68.37, 85.41], Cohen's $d = 4.16$, $BF_{10} > 10^6$; outcome agency: $\Delta M = 41.5\%$, $t(36) = 7.39$, $P < 0.001$, CI 95% [30.13, 52.90], Cohen's $d = 1.64$, $BF_{10} > 10^6$). Furthermore, action-relevant forced-choice blocks were associated with significantly higher choice and outcome agency ratings than action-irrelevant forced-choice blocks (choice agency: $\Delta M = 34.1\%$, $t(36) = 14.88$, $P < 0.001$, CI 95% [29.48, 38.78], Cohen's $d = 1.85$, $BF_{10} > 10^6$; outcome agency: $\Delta M = 24.1\%$, $t(36) = 7.85$, $P < 0.001$, CI 95% [17.91, 30.39], Cohen's $d = 1.03$, $BF_{10} > 10^6$). Our results indicate that the effect of choice condition is larger for choice agency ratings than outcome agency ratings and show that agency experience decreases with a reduction in the freedom to choose between different response options.

Electrophysiological results

Feedback-induced time-frequency oscillations

Topographical plots depict the scalp distribution of theta power 200 to 600 ms after feedback onset (Fig. 4).

Time-frequency maps with midfrontal oscillatory activity during feedback presentation are shown in Fig. 5. Cluster-based permutation analyses revealed a main effect of CHOICE (free-choice/action-relevant forced-choice/action-irrelevant forced-choice) on feedback processing, $P < 0.001$, Cohen's $d = 0.79$. This cluster encompassed the delta and theta range for the entire duration of feedback presentation.

Post-hoc t-tests (two-sided) revealed a significant enhancement in feedback processing for the free-choice condition compared with the action-relevant forced-choice condition (cluster difference: $\Delta M = 0.63$ dB, $SD = 0.25$, $P = 0.001$, Cohen's $d = 0.72$) and for the free-choice condition compared with the action-irrelevant forced-choice condition (cluster difference: $\Delta M = 0.66$ dB, $SD = 0.33$, $P = 0.001$, Cohen's $d = 0.81$; Fig. 6). No significant differences in feedback processing between the action-relevant forced-choice condition and the action-irrelevant forced-choice conditions were detected (largest cluster: $t_{\text{cluster}} = 222.41$, $P = 0.235$). Thus, freedom of choice, but not action relevance, was associated with enhanced neural feedback processing.

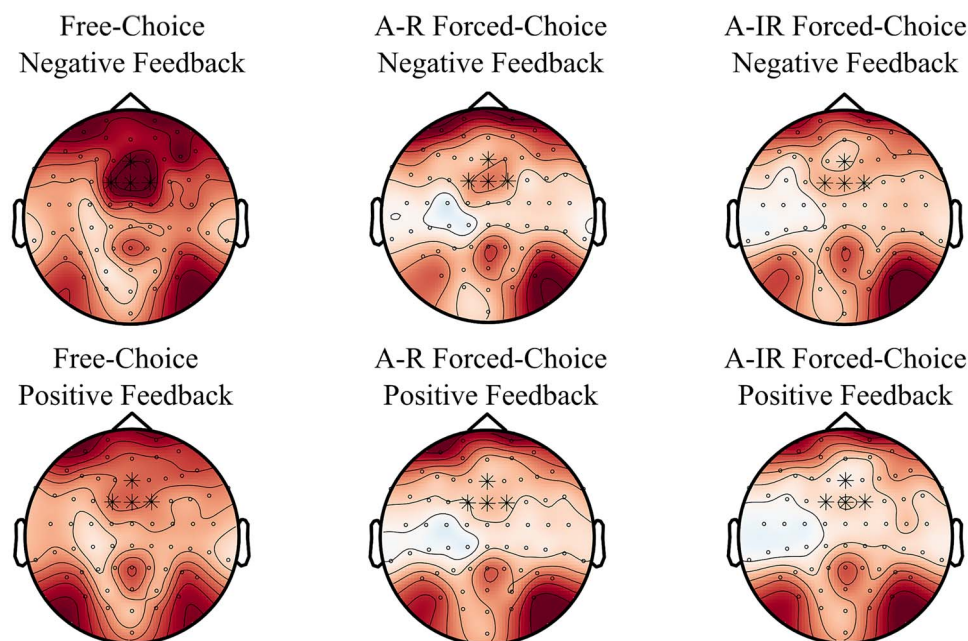


Fig. 4. Topographical plots show the scalp distribution of oscillatory power in the theta range (4 to 7 Hz) between 200 and 600 ms postfeedback onset. Electrodes used for statistical analyses (FC1, FC2, FCz, Fz) are marked as black stars.

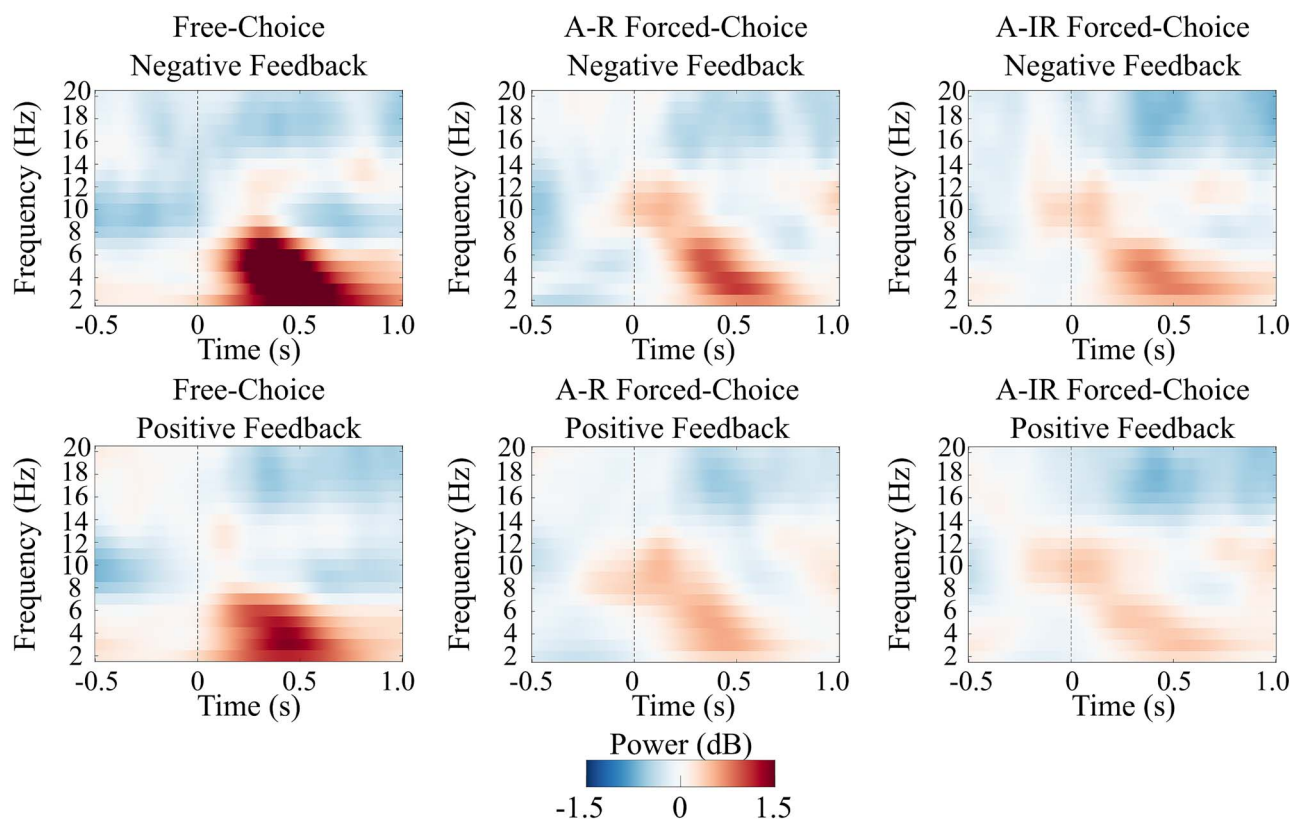


Fig. 5. Time–frequency maps depicting oscillatory power averaged over midfrontal electrodes FC1, FC2, FCz, and Fz for all choice–outcome conditions. Dashed line at time (s) = 0 indicates feedback onset.

We further found a significant effect of VALENCE (negative feedback/positive feedback) on feedback processing, encompassing a cluster from 160 ms until the end of feedback presentation, $P = 0.001$, Cohen's $d = 0.73$. The results indicate that negative feedback led to a significantly larger neural impact than positive feedback (cluster difference: $\Delta M = 0.36$ dB, $SD = 0.16$). As indicated

by follow-up t-tests (two-sided), negative compared to positive feedback led to significantly higher midfrontal oscillatory activity in the free-choice condition (150 to 1,000 ms, Cluster difference: $\Delta M = 0.62$ dB, $SD = 0.29$, $P = 0.001$, Cohen's $d = 0.67$), the action-relevant forced-choice condition (260 to 880 ms, cluster difference: $\Delta M = 0.38$ dB, $SD = 0.08$, $P = 0.004$, Cohen's $d = 0.48$), and the

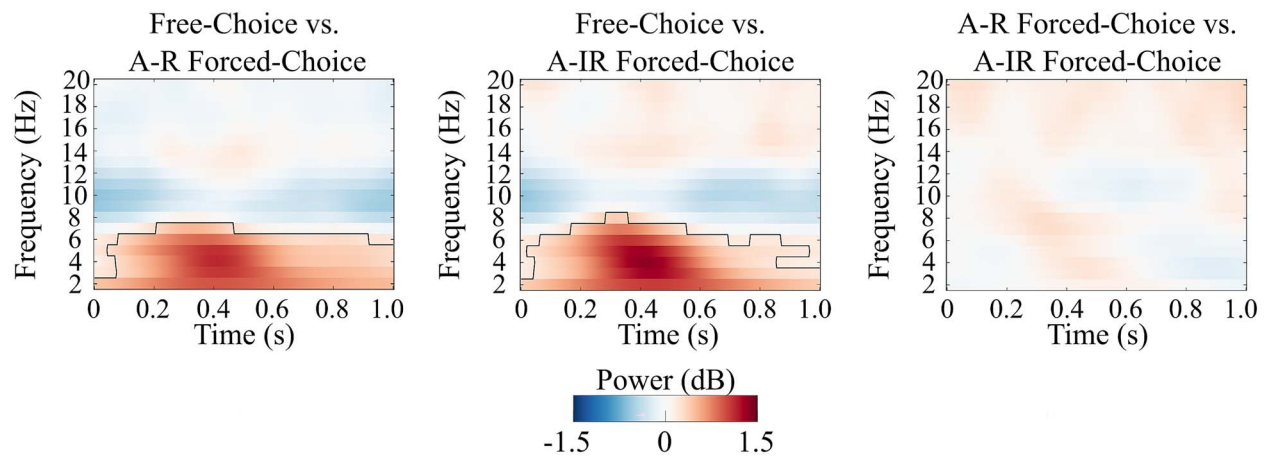


Fig. 6. Results of cluster-based permutation tests, showing the effect of CHOICE (free-choice/action-relevant [A-R] forced-choice/action-irrelevant [A-IR] forced-choice) on feedback processing. Time–frequency points with significant differences between conditions ($P < 0.05$) are marked in black.

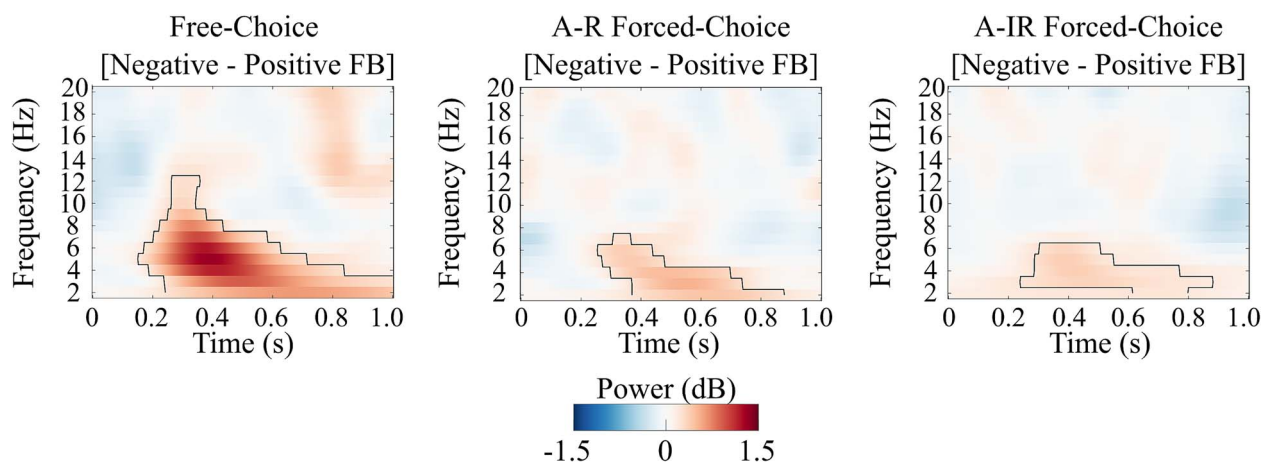


Fig. 7. Results of cluster-based permutation tests, showing the effect VALENCE (negative feedback [FB]/positive feedback [FB]) on feedback processing for free-choice blocks (left), action-relevant [A-R] forced-choice blocks (middle), and action-irrelevant [A-IR] forced-choice blocks (right). Time–frequency points with significant differences between conditions ($P < 0.05$) are marked in black.

action-irrelevant forced-choice condition, 240 to 880 ms, Cluster difference: $\Delta M = 0.28$ dB, $SD = 0.05$, $P = 0.018$, Cohen's $d = 0.41$ (Fig. 7).

A significant 3×2 interaction of the factors CHOICE and VALENCE additionally indicated that the effect of choice on mid-frontal oscillatory power differed for negative and positive feedback, encompassing a cluster from 200 to 600 ms after feedback onset, $P = 0.002$, Cohen's $d = 0.53$. Follow-up t -tests (two-sided) showed that the difference between negative and positive feedback processing was more pronounced for the free-choice condition than for the action-relevant forced-choice condition (180 to 610 ms, cluster difference: $\Delta M = 0.60$ dB, $SD = 0.16$, $P = 0.004$, Cohen's $d = 0.54$; Fig. 8). In addition, free choices, compared to action-irrelevant forced choices elicited significantly larger oscillatory power for negative compared to positive feedback (cluster difference: $\Delta M = 0.54$ dB, $SD = 0.19$, $P = 0.002$, Cohen's $d = 0.48$). In the observed data, this cluster ranged from 200 to 870 ms post feedback onset and encompassed frequencies in the theta and alpha band. No significant differences in feedback-induced mid-frontal oscillatory activity were detected for negative and positive feedback in the action-relevant forced-choice condition compared with the action-irrelevant forced-choice condition (largest cluster: $t_{\text{cluster}} = 140.02$, $P = 0.470$).

In summary, the freedom to choose between different response options enhanced affective feedback processing, as observed in midfrontal low-frequency oscillations in our data, both in comparison to action-relevant and action-irrelevant forced choices. This effect was larger for negative feedback than for positive feedback, indicating a neuronal processing bias for negative feedback. No neural differentiation between action-relevant and action-irrelevant outcomes following forced choices could be observed.

Feedback-evoked event-related potentials

Figure 9 displays individual midfrontal ERP waveforms of all choice-outcome conditions as well as difference waves (positive – negative feedback) of the free-choice condition, action-relevant forced-choice condition, and action-irrelevant forced-choice condition. Scalp distributions of the N100, the RewP nondifference waves, and the RewP difference waves are depicted in Fig. 10.

Cluster-based permutation analyses indicated a significant effect of CHOICE (free-choice/action-relevant forced-choice/action-irrelevant forced-choice) on feedback processing during the first 500 ms of feedback processing, $P < 0.001$, Cohen's $d = 0.75$. T -tests (two-sided) revealed significantly more negative amplitudes during free-choice blocks than action-relevant forced-choice blocks, with a cluster extending from 100 to 140 ms post

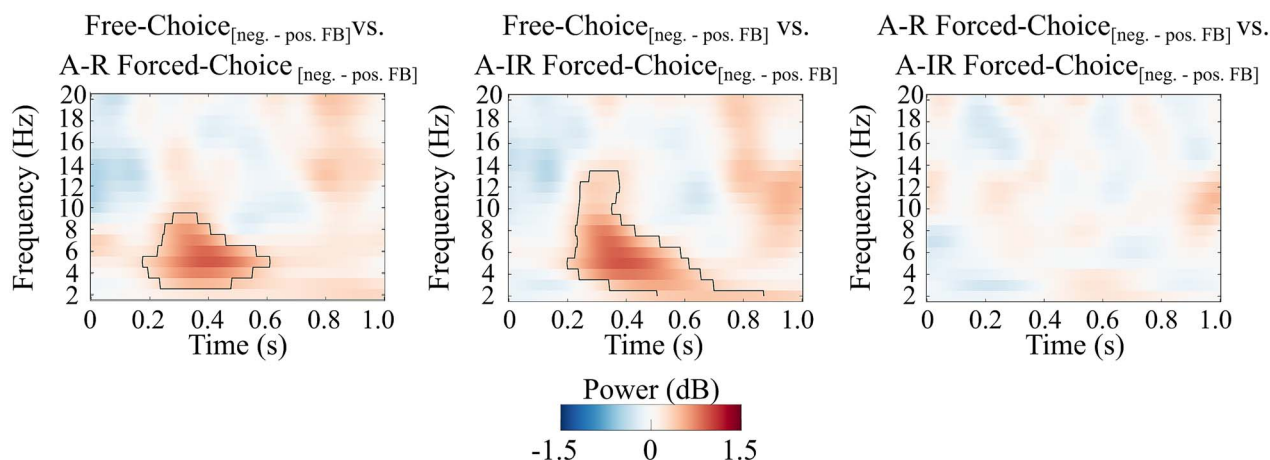


Fig. 8. Results of cluster-based permutation tests, showing the interaction of CHOICE (free-choice/action-relevant [A-R] forced-choice/action-irrelevant [A-IR] forced-choice) and VALENCE (positive feedback [pos. FB]/negative feedback [neg. FB]) over midfrontal electrodes during feedback presentation. Time-frequency points with significant differences ($P < 0.05$) are marked in black.

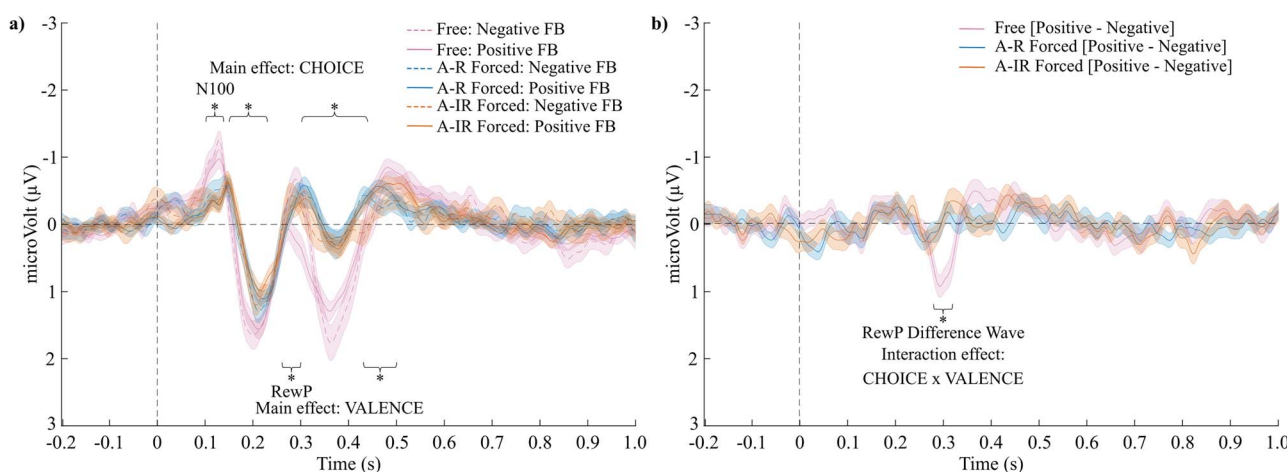


Fig. 9. ERP waveforms over midfrontal scalp sites (FC1, FC2, FCz, Fz). a) Waveforms of all choice (CHOICE: free-choice/action-relevant [A-R] forced-choice/action-irrelevant [A-IR] forced-choice) and outcome (VALENCE: positive feedback/negative feedback [FB]) conditions. Braces with stars above the waveforms indicate time windows for which permutation analyses indicated significantly larger amplitudes in the free-choice compared to the action-relevant and action-irrelevant forced-choice condition. Braces with stars below the waveforms indicate time windows for which permutation analyses indicated significant processing differences for positive and negative feedback. ERP components N100 and RewP are marked. b) Difference waveforms (positive - negative feedback) for the free-choice, the A-R forced-choice, and the A-IR forced-choice condition. The brace with star below the waveforms marks the time window for which a cluster with significant differences was observed in the data. RewP isolated from the difference wave analysis is marked.

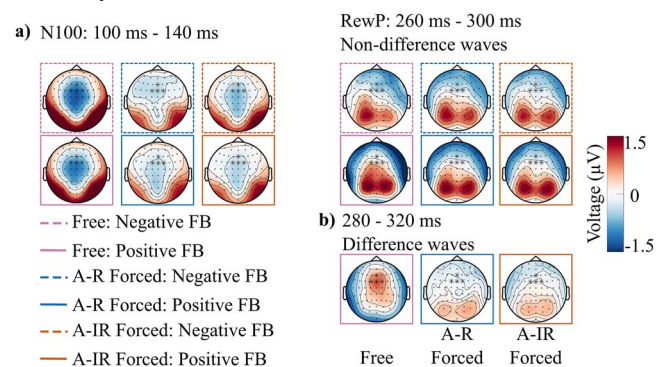


Fig. 10. Topographical plots depicting condition wise scalp distributions of the N100 component (100 to 140 ms), as well as the RewP non-difference (260 to 300 ms) and difference waves (280 to 320 ms) relative to feedback onset. a) Dashed squares around the topographical plots mark negative feedback and solid squares mark positive feedback for the free-choice condition, the action-relevant [A-R] forced-choice condition, and the action-irrelevant [A-IR] forced-choice condition. b) Solid lines are used to mark topographical plots of difference waveforms for the free-choice condition, the A-R forced-choice condition, and the A-IR forced-choice condition.

feedback onset (Cluster difference: $\Delta M = -0.56 \mu V$, $SD = 0.17$, $P = 0.017$, Cohen's $d = 0.70$), indicating an enhanced N100 amplitude for feedback resulting from free choices compared to action-relevant forced choices. We further observed clusters from 150 to 250 ms and 310 to 420 ms postfeedback onset, with permutation tests indicating significantly larger P200 (cluster difference: $\Delta M = 0.60 \mu V$, $SD = 0.18$, $P = 0.002$, Cohen's $d = 0.62$) and P300 amplitudes (cluster difference: $\Delta M = 0.89 \mu V$, $SD = 0.24$, $P < 0.001$, Cohen's $d = 0.83$) in the free-choice condition than the action-relevant forced choice condition during feedback presentation. Free choices also elicited significantly larger amplitudes than action-irrelevant forced choices, with clusters extending over similar time windows (N100: 80 to 140 ms, cluster difference: $\Delta M = -0.44 \mu V$, $SD = 0.15$, $P = 0.012$, Cohen's $d = 0.61$; P200: 150 to 220 ms, cluster difference: $\Delta M = 0.57 \mu V$, $SD = 0.21$, $P = 0.010$, Cohen's $d = 0.60$; P300: 300 to 440 ms, cluster difference: $\Delta M = 0.92 \mu V$, $SD = 0.31$, $P < 0.001$, Cohen's $d = 0.82$). No significant differences in feedback processing could be found for action-relevant and action-irrelevant forced choices during the examined time window (largest cluster: $t_{cluster} = 17.84$, $P = 0.097$). Thus, the freedom to choose between different response options

enhances outcome monitoring, as reflected in larger feedback-evoked N100 amplitudes, as well as later ERPs, compared to both the action-relevant and action-irrelevant forced choices.

Cluster-based permutation tests did not reveal a significant effect of VALENCE (positive feedback/negative feedback) on N100 amplitude (largest cluster: $t_{\text{cluster}} = 6.92$, $P = 0.345$). A cluster with less negative amplitudes for positive compared to negative feedback was observed from 260 to 300 ms post feedback onset (cluster difference: $\Delta M = 0.31 \mu\text{V}$, $SD = 0.06$, $P = 0.025$, Cohen's $d = 0.47$), indicating the presence of an RewP during this time window. In addition, positive feedback elicited significantly more negative amplitudes during later stages of feedback processing (cluster difference: $\Delta M = -0.26 \mu\text{V}$, $SD = 0.05$, $P = 0.002$, Cohen's $d = 0.48$). The observed cluster ranged from 430 to 500 ms after feedback onset.

Finally, a CHOICE $\times$ VALENCE interaction indicated significant differences in RewP amplitude, reflecting processing differences between gain and loss feedback, across choice conditions during the examined time window ($P = 0.001$, Cohen's $d = 0.61$). The detected cluster ranged from 280 to 320 ms after feedback onset. Positive compared to negative feedback elicited significantly larger positive amplitudes in free-choice blocks than in action-relevant forced-choice blocks (cluster difference: $\Delta M = 0.85 \mu\text{V}$, $SD = 0.19$, $P < 0.001$, Cohen's $d = 0.59$) as well as in free-choice blocks than in action-irrelevant forced-choice blocks (cluster difference: $\Delta M = 0.82 \mu\text{V}$, $SD = 0.15$, $P = 0.005$, Cohen's $d = 0.61$). In addition, positive compared to negative feedback evoked significantly more negative amplitudes following free choices than action-irrelevant forced choices, $P = 0.010$, Cohen's $d = 0.53$. This cluster could be observed from 20 to 60 ms after feedback onset (cluster difference: $\Delta M = -0.42 \mu\text{V}$, $SD = 0.08$). No differences in neural processing for positive and negative feedback could be found for action-relevant and action-irrelevant forced-choice blocks (largest cluster: $t_{\text{cluster}} = -19.85$, $P = 0.061$). Post-hoc t-tests (two-sided) confirmed that a RewP was only present for free-choice blocks (cluster difference: $\Delta M = 0.74 \mu\text{V}$, $SD = 0.17$, $P < 0.001$, Cohen's $d = 0.68$), whereas no RewP was observed for either action-relevant (largest cluster: $t_{\text{cluster}} = 0$, $P = 1.0$) or action-irrelevant (largest cluster: $t_{\text{cluster}} = 0$, $P = 1.0$) forced-choice blocks. Additionally, cluster-based permutation analyses for a cluster observed from 420 to 470 ms postfeedback onset indicated a significantly larger negative deflection for negative compared to positive feedback in the action-relevant forced-choice condition (cluster difference: $\Delta M = -0.39 \mu\text{V}$, $SD = 0.05$, $P = 0.014$, Cohen's $d = 0.38$). Also, significantly larger ERP amplitudes could be observed in response to positive compared to negative feedback in the action-irrelevant forced-choice condition, $P = 0.022$, Cohen's $d = 0.45$. In the observed data, this cluster started 10 ms post feedback onset and lasted for 40 ms (cluster difference: $\Delta M = 0.36 \mu\text{V}$, $SD = 0.05$).

In conclusion, freedom of choice enhanced both early and late stages of event-related feedback processing for both positive and negative feedback. While no evidence for an early processing bias for positive or negative feedback could be found (N100), the presence of a RewP during free-choice blocks indicates a processing bias for positive feedback when being endowed with the freedom to choose between different response options.

In response to one reviewer's suggestion to use a less exploratory statistical approach for analyzing the established neurophysiological markers of feedback processing, we also conducted traditional parametric 3×2 ANOVAs and post-hoc t-tests (two-sided) across the time windows, regions, and frequencies of interest. The results from these analyses fully corroborate the findings obtained through cluster-based

permutations. Detailed results of these tests can be found in the [Supplementary Materials](#).

Discussion

We typically perform actions to attain desired outcomes or to prevent the occurrence of undesired ones. The sense of agency influences the ability to adaptively adjust our behavior in accordance with these objectives. To examine how the brain differentiates self- and externally determined actions during goal-directed behavior, the present study examined how choice autonomy influences the processing of affective action outcomes during reinforcement learning. To this end, we recorded EEG while participants completed a computer-based learning task involving free and forced choices.

As intended in reinforcement learning tasks, participants increasingly chose the high-value item throughout block progression during free-choice blocks, thereby maximizing their final payout. Successful learning of item–outcome associations could also be observed in action-relevant forced-choice blocks, reflected in participants' item selections during free choices of this block. Corroborating previous work examining the influence of freedom of choice on learning performance ([Chambon et al. 2020](#)), our results therefore indicate similar feedback learning during free-choice blocks and action-relevant forced-choice blocks, suggesting that participants learned from feedback both during free-choice and forced-choice trials. Furthermore, our results resonate with findings from studies examining learning from action versus observation, showing similar learning rates in both contexts ([Bellebaum et al. 2010](#); [Kobza and Bellebaum 2015](#); [Weismüller et al. 2019](#)). Slower reaction times during free compared to forced choices may further reflect the selective involvement of decision-related processes during self-determined but not externally determined item selections. Supplementary analyses of response-evoked neural processing indicate that this may be reflected in a reduced amplitude of the readiness potential during free compared to forced choices.

Our neurophysiological results show that the freedom to choose between response options enhances both sensory and evaluative markers of feedback processing, reflected in greater N100 amplitudes, larger midfrontal low-frequency power, and a selective RewP during free-choice compared to action-relevant and action-irrelevant forced-choice blocks, respectively. Crucially, prior results showing heightened outcome monitoring for free choices over forced ones (e.g. [Weismüller et al. 2019](#); [Chang et al. 2020](#); [Zheng et al. 2020](#); [Giersiepen et al. 2023](#)) could be attributed to two distinct mechanisms: (i) the association of outcomes with recent self-determined choice, highlighting agency-specific influences on outcome processing, and (ii) the enhanced relevance of outcomes for participants' future decisions, indicating variations in the informational value of feedback across choice conditions. Differing from these studies, our current research discriminates between action-relevant forced-choices (where outcomes are informative for future action decisions) and action-irrelevant forced choices (where outcomes do not inform future decisions), thus enabling a clearer separation of these explanations. If neural feedback processing was sensitive to the relevance of outcomes for future actions, we would have expected enhanced outcome processing for action-relevant compared to action-irrelevant forced choices. Yet, the results from the current study reveal similar feedback-induced outcome monitoring for action-relevant and action-irrelevant forced choices, reflected in both midfrontal oscillatory activity and feedback-evoked potentials, indicating

that these markers do not track the relevance of affective action outcomes for subsequent behavior. Instead, freedom of choice enhances the processing of affective action outcomes, even if forced choice outcomes are relevant for future actions. By demonstrating that free choices enhance outcome monitoring compared to forced choices, our findings corroborate prior research, on observational learning, suggesting that midfrontal outcome processing, signified in MF0 activity, N100, and RewP, may play a predominant role for learning from self- but not externally determined actions (Bellebaum et al. 2012; Kobza et al. 2012; Kobza and Bellebaum 2015). Importantly, by distinguishing between action-relevant and action-irrelevant forced choices, our results indicate that heightened outcome processing during free-choice blocks is driven by the association of outcomes with recent self-determined choice rather than the practical significance of outcomes for future actions. In other words, the enhanced processing of outcomes is more about the psychological impact of making one's own choices, rather than how informative the outcome is for what the individual decides to do next.

Notably, the choice-induced enhancement in outcome processing was not specific to the N100 and RewP but could also be observed in feedback-locked P200 and P300 amplitudes, potentially reflecting increased attentional engagement associated with higher levels of perceived control during free compared to forced choices (e.g. Yeung et al. 2005; Mühlberger et al. 2017; Han et al. 2021). Our results therefore indicate that agency experience, in the form of self-determined choice, leads to a general enhancement in affective outcome monitoring.

At first glance, the enhanced N100 amplitude during self-determined choice may seem contradictory to a large body of work linking agency experience to sensory attenuation (see Gentsch and Schütz-Bosbach 2015). However, considering the functional role of action outcomes, we believe that our findings are consistent with these studies. In studies of sensory attenuation, outcomes typically do not carry instrumental affective value. In contrast, the current study employed motivationally salient action outcomes that inform participants whether item choices were correct (i.e. are likely to result in a gain of money) or incorrect (i.e. are likely to result in a loss of money). Thus, depending on the task context, self-determined choice may selectively increase or decrease the neural impact of action outcomes. In line with this prediction, previous work suggests that N100 amplitude is sensitive to more general processes, including attentional processes and task context (Kaiser and Schütz-Bosbach 2018). Accordingly, our results indicate a greater deployment of attentional resources, reflected in enhanced N100 amplitudes for free- compared to action-relevant and action-irrelevant forced choices.

In line with research associating MF0 activity with conflict monitoring and cognitive control during learning (e.g. Cavanagh et al. 2010), our results showed increased MF0 power for negative feedback compared with positive feedback. Differentiation of gain and loss feedback was observed for all three choice conditions (free-choice/action-relevant forced-choice/action-irrelevant forced-choice), indicating that oscillatory activity in the theta range tracks outcome valence for all self-relevant action outcomes. Yet, this effect was most pronounced in the free-choice condition (see also Zheng et al. 2020), potentially reflecting that negative outcomes are perceived as more aversive when they are self- rather than externally controlled (Majchrowicz and Wierchoń 2021). Negative outcomes under conditions of high agency experience (i.e. free-choice) may consequently induce greater cognitive conflict and elicit more cognitive control to

prevent negative outcomes in the future. Our findings extend previous research comparing active to observational learning (e.g. Weismüller et al. 2019) by suggesting that this cognitive conflict is not solely driven by the instrumental value of the outcomes but rather the self-determination associated with them.

Conversely, analyses of ERP waveforms indicate a valence-specific processing bias for positive feedback. Our results show that an RewP, reflecting increased neural responses for positive compared to negative feedback, was only present in the free-choice condition. Thus, whereas MF0 power was increased for negative feedback compared with positive feedback in both free- and forced-choice conditions, an RewP was present only for self-determined action outcomes. The RewP has been associated with reward evaluation (Becker et al. 2014), reward prediction error (Zheng et al. 2020), motivational intensity, and the informational value of outcomes for subsequent action (Mühlberger et al. 2017). Given similar performance during free-choice blocks and action-relevant forced-choice blocks as well as the implementation of identical item–outcome associations in all choice conditions, differences in the instrumental value of feedback or outcome predictability unlikely account for the selective presence of a RewP during free-choice blocks. Rather, the motivational salience or subjective value of outcomes seems to be increased when actions are self- rather than externally determined (see also Mei et al. 2018). Accordingly, recent behavioral results indicate that only free choices are associated with differential weighting of positive and negative feedback during reinforcement learning, whereas outcomes of forced choices are treated impartially, meaning valence-neutral, regardless of their positive or negative affective value (Chambon et al. 2019, 2020).

Interestingly, while many previous studies have observed a diminished RewP for positive outcomes after forced choices compared to free choices (e.g. Yeung et al. 2005; Mühlberger et al. 2017; Mei et al. 2018; Hassall et al. 2019; Zheng et al. 2020), these studies typically employed gambling tasks with random reward schedules. Consequently, while participants may use the outcomes of their actions to influence their subsequent choices, the randomness limits their ability to establish consistent item–outcome associations for reliable outcome prediction over time. In contrast, the reinforcement learning task of the current study enabled participants to develop concrete outcome expectations, thereby diminishing the element of surprise (i.e. prediction error) associated with receiving feedback. Considering that RewP amplitude is known to decrease with increasing outcome predictability (Krigolson et al. 2014; Williams et al. 2020; Weber and Bellebaum 2024), the absence of a RewP in both action-relevant and action-irrelevant forced choices in the current experiment suggests that this neural marker differentiates between positive and negative outcomes based on their predictability from prior actions, but only when those outcomes were self-, not externally, controlled. In the current experiment, successful learning was only established based on free choices in the free-choice and action-relevant forced-choice blocks. This approach was necessary, since including an assessment of learning in the action-irrelevant forced-choice blocks would have made the feedback automatically action-relevant. Nevertheless, we cannot fully exclude the possibility that differences in learning-related reward prediction errors between free choices, action-relevant forced-choices, and action-irrelevant forced choices influenced the observed differences in outcome processing. If outcome processing in the current study was modulated by reward prediction error, the absence of significant processing differences between the action-relevant and action-irrelevant

forced-choice condition could be interpreted as an indicator of comparable learning in both conditions. However, this possibility needs to be further tested in future studies.

Taken together, both the increased impact of negative feedback on MF0 oscillations and positive feedback on RewP during free choices may be reconciled within the reward-based framework of perceived control (Leotti et al. 2015; Murayama et al. 2015; Ly et al. 2019; Wang et al. 2021). The framework builds on theories of perceived control and empirical evidence that both highlight the fundamental drive of humans to exert control when interacting with the environment. More specifically, the opportunity to choose is thought to be inherently rewarding and to elicit activity in reward-related dopaminergic cortico-striatal circuits (Fujiwara et al. 2013). In addition, activity in the medial prefrontal cortex (including the ACC) in response to aversive events has been found to be modulated by perceived control (Kolling et al. 2016), pointing toward improved coping with aversive events when endowed with the opportunity to exert control through choice. Consequently, freedom of choice is suggested to increase motivational and affective processing in response to both positive and negative feedback. The choice-induced enhancement in MF0 power and RewP amplitude in response to negative and positive feedback, respectively, therefore plausibly reflects an increase in the motivational salience and subjective value of affective action outcomes when they are self-compared to externally determined (see also Gentsch et al. 2015; Majchrowicz and Wierchoń 2021; Wang et al. 2021). In contrast to MF0 oscillations and the RewP, N100 amplitude was unaffected by feedback valence. The N100 reflects early stages of feedback processing, and its amplitude has been shown to be modulated by selective attention and the predictability of action outcomes (Hillyard et al. 1998; Kaiser and Schütz-Bosbach 2018). In line with the interpretation that freedom of choice increases motivational processing, the increase in N100 amplitude for free compared to forced choices might reflect an increased deployment of attentional resources to monitor affective action outcomes. In a similar vein, sense of agency over action outcomes has previously been reported to increase N100 amplitude in response to auditory effects without affective valence, if outcomes under conditions of high and low agency experience are equally predictable (Kaiser and Schütz-Bosbach 2018).

To examine whether our freedom of choice manipulation also affected participants' agency judgments, we complemented our neurophysiological measures with self-reported agency ratings after each block. As expected, participants reported highest choice agency ("I was free to choose any item I wanted in this block.") during free-choice blocks, followed by action-relevant forced-choice blocks and action-irrelevant forced-choice blocks. A similar albeit smaller decrease from free-choice to action-irrelevant forced-choice blocks could be observed for outcome agency ("I was in control over the amount of money I gained in this block.") ratings. Thus, participants' agency experience increased when they were endowed with the freedom to choose between different response options. These findings highlight that enhanced outcome processing during self-determined choice is paralleled by greater perceived control during reinforcement learning. When experiencing control, we may be more inclined to monitor the opportunities for interaction available in our environment, resulting in greater sensitivity, especially for motivationally salient stimuli that indicate our success or failure of achieving a desired goal. Increased outcome monitoring in situations of high perceived control may consequently reflect an increased investment of cognitive resources, such as cognitive control and conflict monitoring (Frömer et al. 2021).

Retrospective agency ratings after each block were in line with the ratio of free-choice trials of that block, with lowest agency ratings for blocks with only forced-choice trials (action-irrelevant forced choice), higher agency ratings with blocks containing some free-choice trials (action-relevant forced choice), and highest agency ratings for blocks with only free choices (free choice).

Notably, the differential effect of block type on choice and outcome agency ratings highlights the multifaceted nature of the sense of agency, shaped by distinct cues to agency experience (e.g. Synofzik et al. 2013; Sidarus et al. 2017; Hassall et al. 2019). Whereas choice comprised a reliable, deterministic cue to the sense of agency in the current experiment, outcomes were probabilistically associated with item choices. Accordingly, our agency manipulation more strongly affected participants' freedom of choice judgments than their perceived control over action outcomes. These results are in line with predictions from optimal cue integration accounts of the sense of agency (Synofzik et al. 2013) and support previous findings that illustrate the influence of distinct task components on the sense of agency (Giersiepen et al. 2023).

As self-reported agency experience was assessed after participants completed individual blocks of trials, it should be noted that their ratings may be strongly shaped by retrospective processes, including the predictability of action outcomes and a higher rate of positive compared to negative feedback during learning blocks (Synofzik et al. 2013). Consequently, agency ratings are inclined to reflect higher-level beliefs rather than moment-to-moment changes in agency experience, precluding the assessment of immediate, lower-level changes in the sense of agency such as the influence of outcome valence. Likewise, future studies could investigate the relation between subjective agency ratings and neural reactivity during feedback presentation more closely by having participants rate their agency after each trial. Including trial-wise agency judgments in future studies using motivationally salient positive and negative feedback would advance our understanding of how affective processes influence agency experience and outcome evaluation.

Conclusion

In conclusion, our finding that freedom of choice during reinforcement learning enhances MF0 oscillations as well as early (N100) and late (RewP) event-related markers of feedback processing, irrespective of the relevance of forced-choice outcomes for subsequent actions, suggests that linking outcomes to self-determined decisions heightens their motivational importance and perceived value. However, further research concerning the functional role of choice-induced effects on affective outcome processing is needed to develop a comprehensive understanding of the practical significance of these results. During everyday behavior, the affective valence of outcomes is central for assessing actions as successful or not. In a recent review, Kaiser et al. (2021b) proposed that outcome processing functionally mediates agency-related improvements in action regulation. Whether increased outcome monitoring under conditions of self-determined choice leads to improved behavioral adaptation should therefore be directly tested in future studies. In a similar vein, further studies are needed to systematically test whether increased MF0 power and RewP amplitudes during free choices are predominantly driven by the motivational salience of action outcomes, by an increase in the subjective value of self-determined action outcomes, irrespective of motivational involvement, or whether these two processes

interact in the neural enhancement of self-determined action outcomes.

To summarize, our study shows that the sense of agency, operationalized as freedom of choice, enhances the neural impact of positive and negative feedback during reinforcement learning, as reflected in an enhanced N100 and RewP amplitude as well as increased MF0 power for free compared to forced choices. Crucially, we show that this choice-related enhancement in outcome processing is not driven by the relevance of outcomes for future actions but rather stems from the association of outcomes with self-determined choice. Taking into consideration the reward-based framework of perceived control, we interpret more pronounced theta power for negative feedback and the selective expression of an RewP during free choices as indicative of increased motivational salience and subjective value of self-compared to externally induced action outcomes. Future research is required to further disentangle motivational processes and agency-related differences in subjective value and test the functional significance of enhanced outcome processing during self-determined choice.

Acknowledgments

We thank Elif Gezen, Jakob Brunner, Lea Zaric, Yagmur Karakoc, and Yannick Janvier for their help with data collection.

Author contributions

Maren Giersiepen (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Visualization, Writing—original draft), Simone Schütz-Bosbach (Conceptualization, Funding acquisition, Supervision, Writing—review & editing), and Jakob Kaiser (Conceptualization, Funding acquisition, Methodology, Software, Supervision, Writing—review & editing).

Supplementary material

Supplementary material is available at *Cerebral Cortex* online.

Funding

This work was supported by Deutsche Forschungsgemeinschaft (SCHU 2471/7–1 to S.S.-B. and KA 5468/1–1 to J.K.).

Conflict of interest statement: None declared.

Data availability

Data and study materials can be found online at https://osf.io/6tsxv/?view_only=fd79377cfc548e29155337e665a4ff5.

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